

```
~/workshop $ ./start.sh
```

# Deploy Pertamamu ke Kubernetes.

Dari nol sampai pipeline jalan — di VPS-mu sendiri.

// TODAY'S LINEUP

# Yang bakal nemenin kamu deploy hari ini.

// SPEAKER

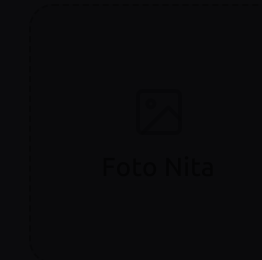
**Ian Yulianto**

Senior DevOps Engineer

LinkedIn

LINKEDIN /in/ianyulianto

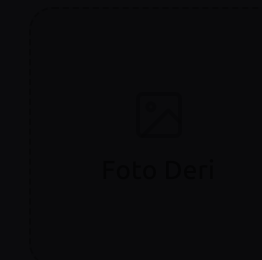
// FASILITATOR



**Nita**

DevOps Engineer

IN /in/nita-oktaviani



**Deri**

DevOps Enthusiast

IN /in/deri-nugroho

// stuck pas hands-on? raise hand – fasilitator  
standby.

# Apa yang akan kita lakukan dalam **3 jam** ke depan?

01

~ 1 JAM

## Theory

Konsep DevOps, containers, Git, CI/CD, Kubernetes — semua fundamental yang kamu butuhin sebelum nyentuh terminal.

02

~ 2 JAM

## Hands-on

SSH → K3s install → Kubeconfig → GitHub Actions → Push → Pipeline jalan. 7 step. Live.

03

// OUTCOME

## Kamu pulang dengan

Pipeline auto-deploy beneran jalan di VPS-mu sendiri. Push code → app update. Itu beneran.

# DevOps bukan jabatan. Ini cara kerja.

01

## Kolaborasi Dev & Ops

Tim yang ngoding dan tim yang nge-run server gak boleh saling lempar masalah.

02

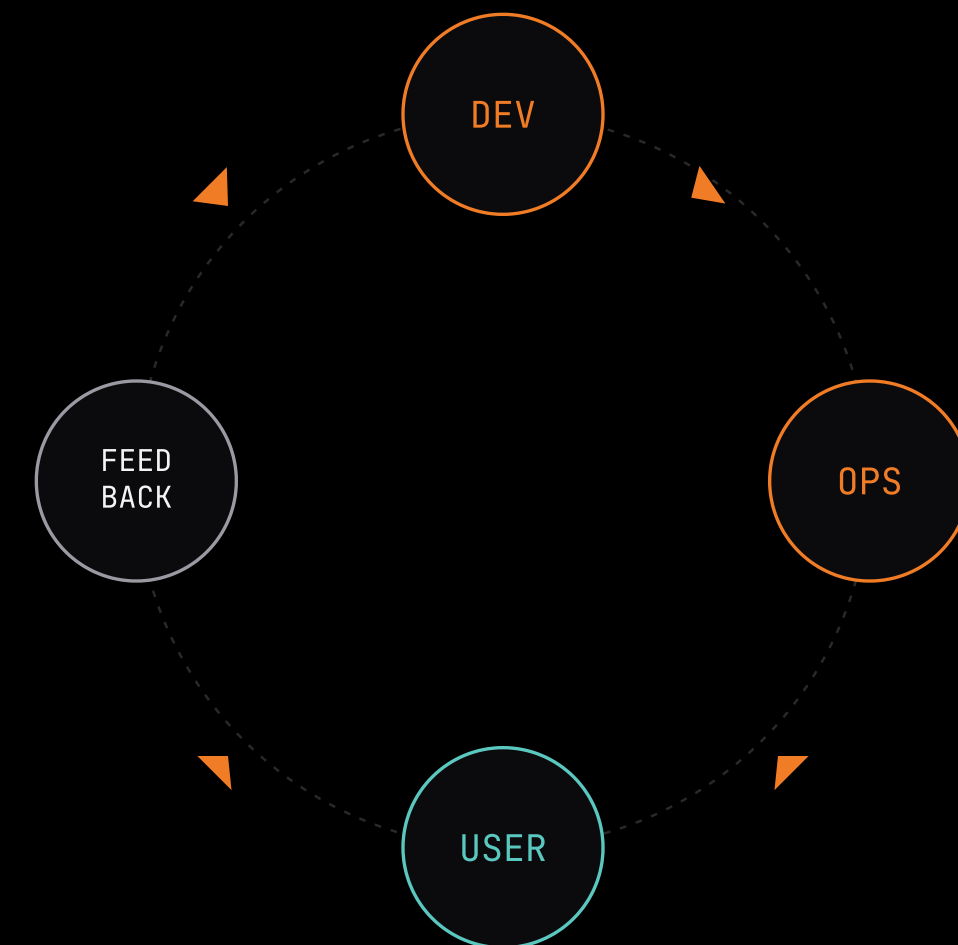
## Feedback loop cepat

Bug ketahuan dalam menit, bukan minggu. Push → test → tau hasilnya.

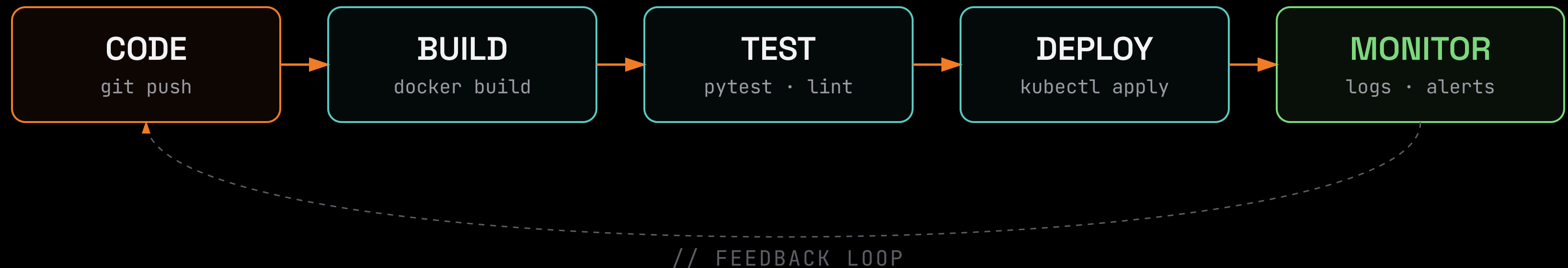
03

## Deliver value sering, aman

Release tiap hari bukan tiap kuartal. Kecil, sering, dengan safety net.



# Dari commit ke production — ini jalurnya.



Setiap stage bisa di-automate. Tujuan kita hari ini: **jalanin loop ini end-to-end.**



# Laptop beda. Server beda. App tetap sama.

## // DEV LAPTOP

macOS 14 • arm64  
node 22.3 • python 3.12  
redis 7.0 • postgres 16

 your-app:v1

✓ works on my machine



## // PROD SERVER

ubuntu 22.04 • x86\_64  
no node • no python  
no redis • no postgres

 your-app:v1

✓ works on the server too

### → portable

Jalan di mana aja yang punya Docker.

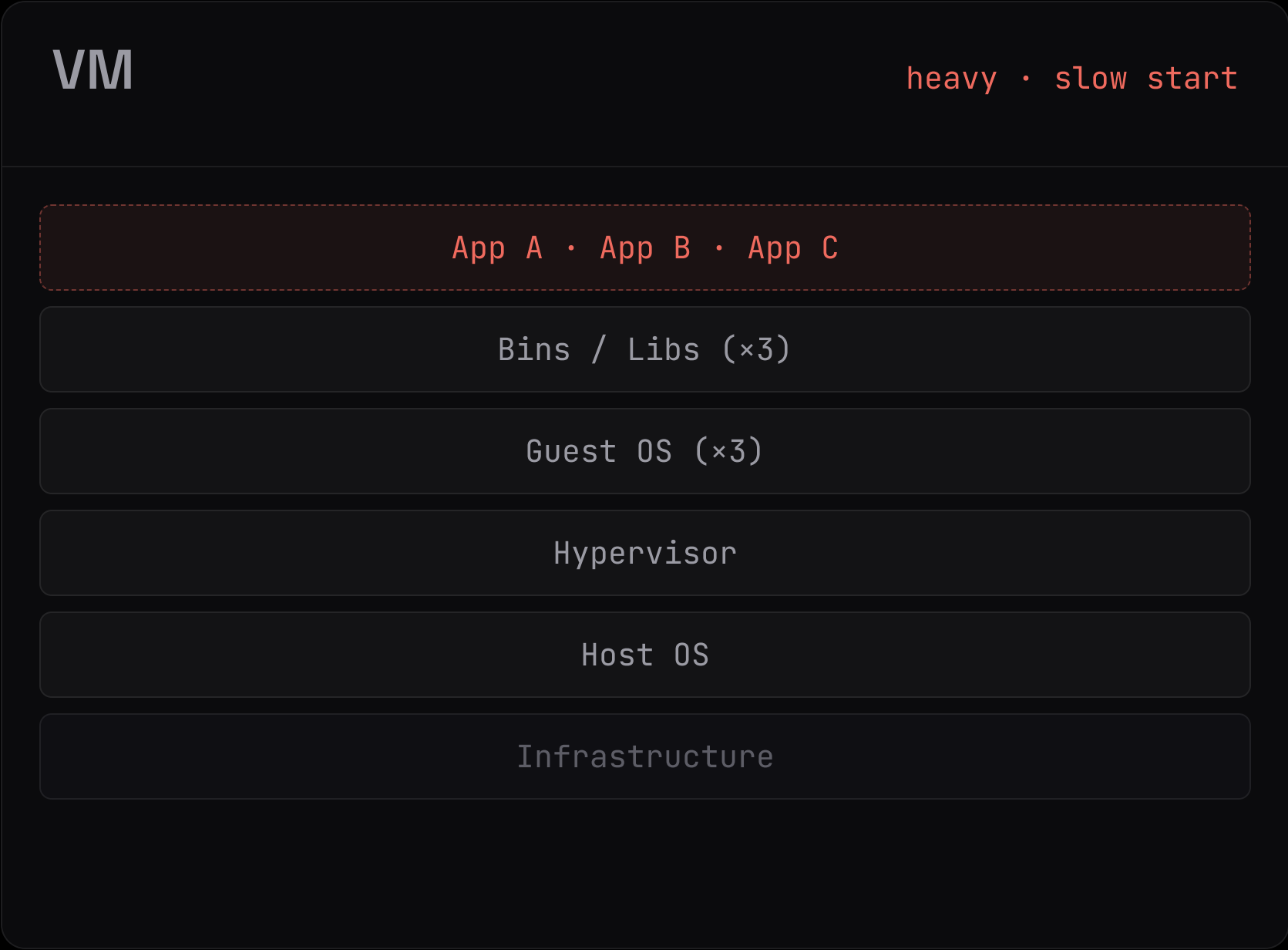
### → isolated

App-mu gak bentrok sama app sebelah.

### → reproducible

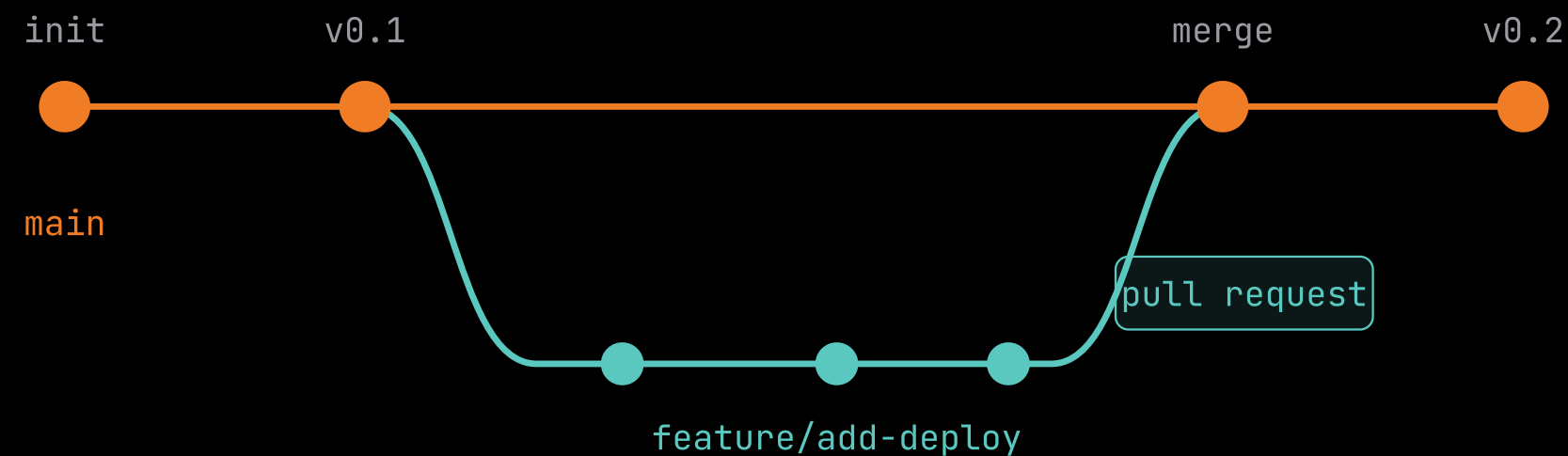
Build sekarang, build besok — hasilnya sama.

# Container vs VM — cukup satu slide.



# Git bukan **Dropbox**.

Git nyimpen sejarah, bukan cuma file terbaru.  
Itu yang bikin kolaborasi & pipeline bisa jalan.



## 01 • VERSION CONTROL

Tau apa berubah, kapan, kenapa.

## 02 • COLLABORATION

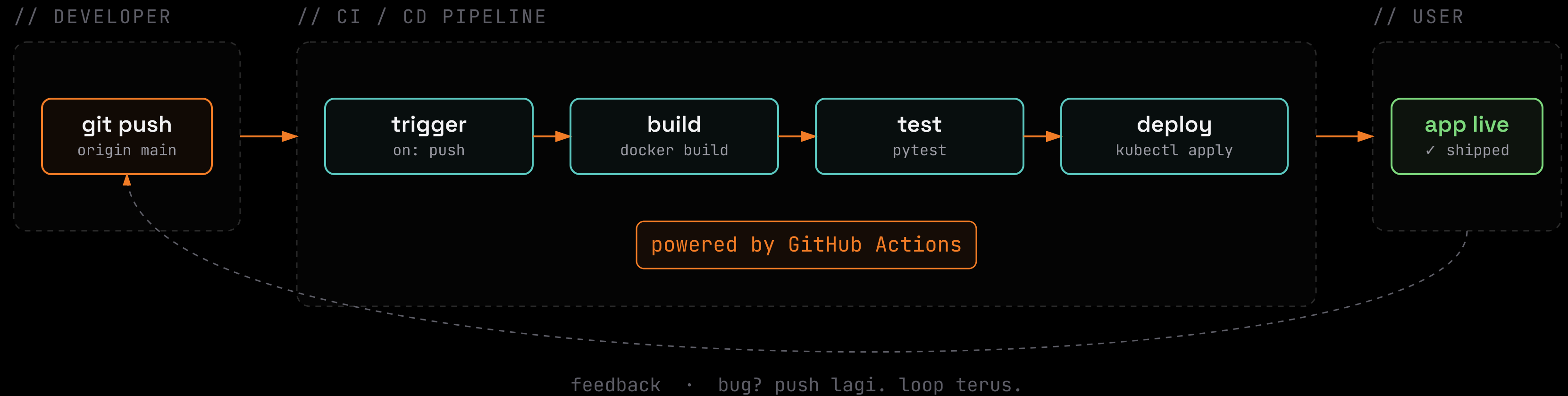
Banyak orang, satu codebase, tanpa nimpa.

## 03 • TRIGGER

**git push** = sinyal buat pipeline.



# Supaya deploy gak **manual selamanya**.



# Kubernetes itu apa?

## Dan kenapa kamu harus tahu.

### POD

**Unit terkecil — bungkus 1+ container.**

Pod mati? K8s bikin baru. Otomatis.

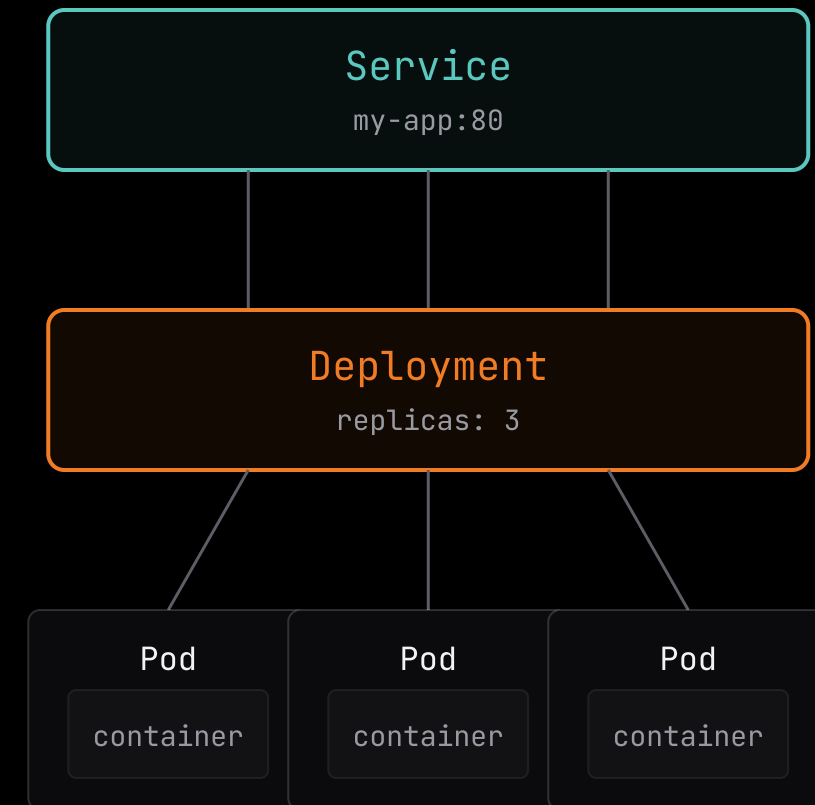
### DEPLOYMENT

**Aturan "harus ada N pod jalan".**

Mau update? Rolling update — gak ada downtime.

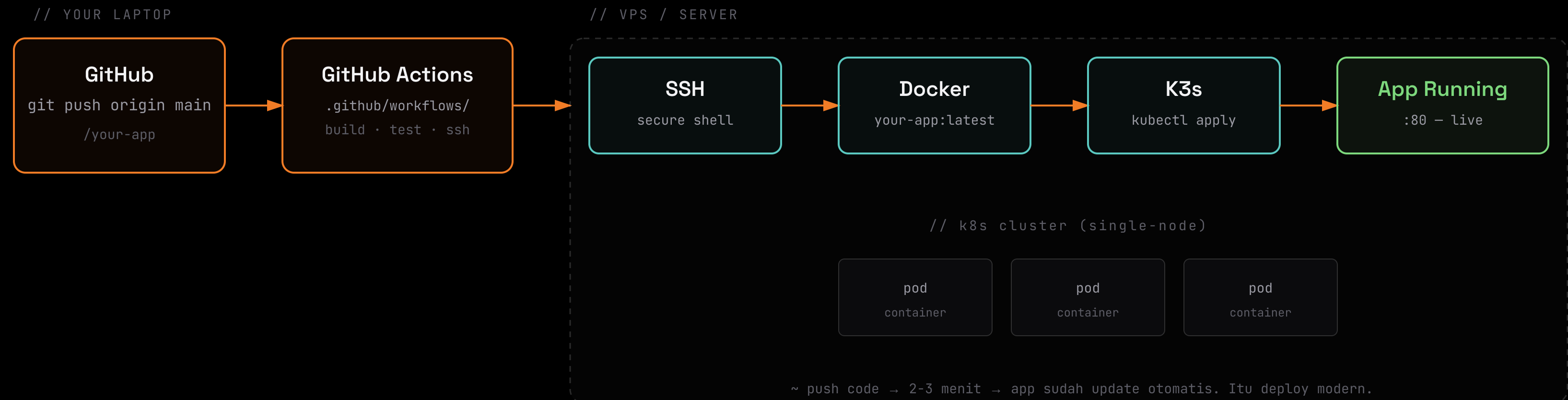
### SERVICE

**Pintu masuk — alamat tetap untuk pod yang berubah-ubah.**



satu pod mati → k8s spawn baru → service tetap hidup

# Ini yang akan kita bangun.



# Tools yang kita pakai hari ini.

// CI/CD

## GitHub Actions

Trigger pipeline tiap kali kamu **git push**.  
Gratis untuk repo public.

// ORKESTRATOR

## K3s

Kubernetes versi ringan. Cocok untuk VPS  
kecil & belajar. 1 command install.

// CONTAINER

## Docker

Bungkus app + dependency jadi satu  
image yang jalan di mana aja.

// gratis • open source • production-ready

# Gimana pipeline GitHub Actions **bekerja?**

```
name: deploy
on:
  push:
    branches: [main]

jobs:
  deploy:
    runs-on: ubuntu-latest
    steps:
      - uses: actions/checkout@v4
      - name: Deploy to K3s
        env:
          KUBECONFIG_DATA: ${ secrets.KUBECONFIG }
        run: |
          kubectl apply -f k8s/
```

→ **on:**

Kapan pipeline trigger. **push** ke branch **main**.

→ **jobs:**

Container of steps. Tiap job jalan di mesin sendiri.

→ **steps:**

Command yang dijalankan satu per satu. Top-down.



# K3s = Kubernetes yang gak bikin pusing.

| ASPEK    | Kubernetes Full                       | K3s                                          |
|----------|---------------------------------------|----------------------------------------------|
| resource | ~2 GB RAM minimum, banyak komponen    | < 512 MB RAM, 1 binary                       |
| setup    | kubeadm / kops, ribet, multi-step     | curl ...   sh -                              |
| use case | Production skala besar, multi-cluster | Dev, lab, edge, VPS kecil — & kita hari ini. |
| api      | 100% kubectl, semua resource          | 100% sama. kubectl jalan persis sama.        |

// tl;dr — K3s ringan, tapi yang kamu pelajari hari ini = transferable ke production K8s.

// THEORY.EXIT()

# Cukup teori. Waktunya jalan.

Buka terminal kamu.

```
~/workshop - bash  
  
$ ssh workshop@vps  
# password: ada di Discord  
$ █
```

STEP 1 / 7 // SSH KE CONTAINER KAMU

## >\_ Connect ke VPS-mu.

Buka terminal. Ketik command di bawah. Ganti placeholder dengan credential kamu (cek Discord).

~/local - bash

```
$ ssh [username]@[ip-address] -p [port]
```

💡 credential lengkap (username • ip • port • password)  
ada di channel #workshop di Discord.



# >\_ Ambil kodenya.

```
[username]@vps - bash

$ git clone [repo-url]
$ cd [repo-name]
$ ls
```

// output yang kamu lihat:

|                  |                          |
|------------------|--------------------------|
| 📁 app/           | ← source kode            |
| 📁 k8s/           | ← manifest yaml          |
| 📁 .github/       | ← workflows              |
| 📄 install-k3s.sh | ← installer script       |
| 📄 Dockerfile     | ← container recipe       |
| 📄 README.md      | ← baca dulu kalo bingung |

STEP 3 / 7 // INSTALL K3S SINGLE-NODE

# > \_ Pasang **Kubernetes**-nya.

[username]@vps - bash

```
$ bash install-k3s.sh
```

|                                   |                        |      |
|-----------------------------------|------------------------|------|
| [INFO] Downloading k3s binary     | <div><div></div></div> | 100% |
| [INFO] Installing systemd service | <div><div></div></div> | 100% |
| [INFO] Starting k3s.service       | <div><div></div></div> | 100% |
| [INFO] Waiting for node Ready     | <div><div></div></div> | 72%  |

🕒 estimasi: 2-3 menit. Jangan close terminal. Kalau muncul node Ready → lanjut step 4.

**STEP 4 / 7** // SETUP KUBECONFIG SEBAGAI GITHUB SECRET

# >\_ Kasih GitHub akses ke cluster-mu.

a) ambil isi kubeconfig

```
[username]@vps - bash

$ sudo cat /etc/rancher/\
  k3s/k3s.yaml

# copy seluruh output
```

b) paste ke GitHub Secret

- ▶ buka repo di GitHub
- ▶ Settings → Secrets and variables
- ▶ → Actions → New repository secret
- ▶ Name: KUBECONFIG
- ▶ Secret: (paste output di sini)
- ▶ Add secret ✓

⚠ Jangan commit file ini ke repo. Secret ≠ public. Ever.

STEP 5 / 7 // TRIGGER PIPELINE PERTAMAMU

# >\_ Push. Magic happens.

```
... [username]@vps - bash

$ git add .
$ git commit -m "trigger deploy"
$ git push origin main
```

// terus apa?

1. buka [github.com/your-repo](https://github.com/your-repo)
2. klik tab **Actions**
3. liat workflow run terbaru
4. klik buat liat log real-time
5. tunggu sampe ✓ hijau

```
trigger → build → ssh + apply → ✓ deployed
```

~ 90 detik

STEP 6 / 7 // PASTIIN APP BENERAN JALAN

> \_ Verify it's **alive**.

[username]@vps - bash

\$ kubectl get pods

| NAME                 | READY | STATUS  | RESTARTS | AGE |
|----------------------|-------|---------|----------|-----|
| your-app-7d4f8c-abcd | 1/1   | Running | 0        | 42s |

\$ kubectl get svc

| NAME     | TYPE         | CLUSTER-IP | EXTERNAL-IP   | PORT(S)      | AGE |
|----------|--------------|------------|---------------|--------------|-----|
| your-app | LoadBalancer | 10.43.0.42 | <your-vps-ip> | 80:31234/TCP | 42s |

**Kalau STATUS = Running → kamu berhasil!**Buka <http://<your-vps-ip>> di browser. App-mu di sana.



**BONUS STEP** // WHAT IF IT BREAKS?

# >\_ Bunuh container-nya. Liat apa terjadi.

// 01 – break

```
$ kubectl delete pod \
  your-app-7d4f8c-abcd
```

// 02 – k3s react

```
# otomatis:
  spawn new pod
  attach service
  healthy in ~5s
```

// 03 – restore

```
$ kubectl get pods
# NAME ...
# your-app-new-xyz
# Running
```

**Ini lah self-healing.** Pod mati → K3s detect → spawn baru → service kembali. Tanpa kamu ngapa-ngapain.

# Tadi kamu **ngapain** aja?

- |   |            |                                        |                     |
|---|------------|----------------------------------------|---------------------|
| ✓ | step<br>01 | SSH ke VPS                             | ssh user@ip -p port |
| ✓ | step<br>02 | Clone repo workshop                    | git clone ...       |
| ✓ | step<br>03 | Install K3s single-node                | bash install-k3s.sh |
| ✓ | step<br>04 | Setup kubeconfig sebagai GitHub Secret | KUBECONFIG          |
| ✓ | step<br>05 | Trigger pipeline GitHub Actions        | git push            |
| ✓ | step<br>06 | Verify pod Running + app live          | kubectl get pods    |
| ★ | bonus      | Self-healing – break & restore         | kubectl delete pod  |

// WHAT'S NEXT?

# Kamu mau **ke mana** setelah hari ini?

// PATH\_01

## Belajar mandiri

Kurikulum lanjutan, cheat sheet, dan resource pilihan. Semua di channel **#resources** Discord.

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→ [discord.gg/DUKXgcwnFG](https://discord.gg/DUKXgcwnFG)

// PATH\_02 — FLAGSHIP

## CBC Cohort

Program lanjutan Debug Life Indonesia. Cohort-Based Course untuk bawa kamu dari "tau" ke "bisa ngerjain".

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→ **coming soon**

// PATH\_03

## Debug Challenge

Tantangan mingguan komunitas. Kompetitif, fun, dan bikin GitHub commit graph kamu ijo terus.

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→ tiap senin di Discord

Gabung komunitas DLI — **perjalanan baru dimulai dari sini.**



# Terima kasih sudah *hadir*.

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JOIN THE COMMUNITY

[discord.gg/DUkXgcwnFG](https://discord.gg/DUkXgcwnFG)

OUR MOTTO

## Inspect • Reflect • Refactor Our Life.